Shihab Ahmed

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Academic Qualifications

University of California, Merced — **Merced, CA, USA**
Ph.D. in Mechanical Engineering — [Jan 2023 – Present]
Advanced to Ph.D. Candidacy: November 2025
CGPA: 3.84/4.00

University of California, Merced — **Merced, CA, USA**
M.S. in Mechanical Engineering — [Conferred May 2026]

Bangladesh University of Engineering and Technology (BUET) — **Dhaka, Bangladesh**
B.Sc. in Mechanical Engineering — [Jul 2014 – Oct 2018]
CGPA: 3.51/4.00

Research Experience

Graduate Student Researcher
Martini Research Group, University of California, Merced

Antioxidant Modeling and Molecular Design — Jan 2023 – Present
Collaborator: Austrian Competence Center for Tribology (AC2T)

- Developing generative AI modeling frameworks based on GANs and diffusion models to design novel lubricant antioxidant molecules with improved performance.
- Performed large-scale reactive molecular dynamics simulations on thousands of candidate molecules to quantify oxidation mechanisms, hydrogen-transfer kinetics, and radical stability.
- Designed and deployed automated high-throughput MD workflows using Python and Bash for scalable execution, monitoring, and data aggregation on SLURM-based HPC clusters.
- Built a literature-mining pipeline using large language models deployed on GPU-enabled HPC nodes to extract and curate lubricant-relevant antioxidant compounds from scientific publications.

MLIP Modeling of Lubricant Additive Interactions — Oct 2026 – Present
Collaborator: Chevron Oronite

- Investigating ZDDP–ADPA interactions using machine-learning interatomic potentials (MLIPs), with a focus on understanding the molecular origins of synergy and antagonism between lubricant additives.

Mentoring

Mentor, Undergraduate Researcher — Summer 2026 – Present
Martini Research Group, University of California, Merced

- Mentoring an undergraduate researcher in applied machine learning, covering overfitting diagnosis (train-test gap, learning curves, k-fold cross-validation), model comparison across random forest, multilayer perceptron, and support vector machine approaches, and regularization concepts.
- Guiding the student in research workflow and independent problem-solving alongside technical instruction.

Publications

Journal Publications

- Ahmed S, Eder SJ, Iqbal MM, Dörr N, Martini A (2026) "Reactive MD screening of antioxidants for substituent-dependent phenoxyl radical stability", *ACS Omega*, **11**, 16886. DOI: 10.1021/acsomega.6c00592.
- Ahmed S, Eder SJ, Dörr N, Martini A (2024) "Tracking thermo-oxidation reaction products and pathways of modified lignin structures from reactive molecular dynamics simulations", *Journal of Physical Chemistry A*, **128**, 5398–5407. DOI: 10.1021/acs.jpca.4c00964.

Oral Presentations

- (Accepted) *Generative AI and Reactive Molecular Dynamics-Guided Discovery of Phenolic Lubricant Antioxidants*, ÖTG Symposium, Austrian Tribology Society, Vienna, Austria, November 2026
- *Machine Learning Prediction of Antioxidant Additive Performance from Atomistic Simulations*, TMS Annual Meeting & Exhibition, San Diego, CA, March 2026
- *Reactive MD screening of antioxidants for substituent-dependent phenoxyl radical stability*, ME Graduate Symposium, University of California, Merced, February 2026.
- *"Understanding the Mechanisms of Antioxidants Using Reactive Molecular Dynamics"*, ACS Spring 2025 National Meeting & Exposition, San Diego, CA, March 2025
- *"Understanding the Mechanism of Antioxidant Additives Using Reactive Molecular Dynamics",* ME Graduate Symposium, University of California, Merced, November 2024.
- *"Understanding the Thermo-Oxidation Pathways of Antioxidants Using Reactive Molecular Dynamics",* Gordon Research Seminar on Tribology, Lewiston, ME, June 2024.
- *"Fabrication of a Cost-Effective Prosthetic Arm Using Electroencephalography Signal",* International Conference on Industrial & Mechanical Engineering and Operations Management, Dhaka, Bangladesh, December, 2020.

Poster Presentations

- *"Understanding the Thermo-Oxidation Pathways of Antioxidants Using Reactive Molecular Dynamics,"* Gordon Research Conference on Tribology, Lewiston, ME, June 2024.

Professional Service

Peer Reviewer 2025 – Present

- **ASME Journal of Tribology**: reviewed three manuscripts on molecular simulation of lubricant oxidation and sustainable bio-based lubricants.
- **Tribology Letters**: reviewed one manuscript on machine learning prediction of base oil tribological properties from molecular dynamics simulations.

Professional Experience

University of California, Merced **Merced, CA, USA**
Graduate Student Researcher Jan 2023 – Present

- Researching eco-friendly oxidation inhibitors for lubricants using molecular dynamics simulations.
- Investigating reaction pathways and oxidation resistance to support research publications.

Bangladesh-India Friendship Power Company Limited (BIFPCL) **Khulna, Bangladesh**
Assistant Manager, O&M Commissioning (Mechanical) Dec 2020 – Jul 2022

- Attended comprehensive training on operations and maintenance (O&M) best practices at the Power Management Institute in India, gaining advanced technical skills and operational expertise.

Daffodil International University **Dhaka, Bangladesh**
Part-Time Lecturer May 2019 – Apr 2020

- Taught foundational courses in mechanical engineering, including theory of machines, solid mechanics, and fluid mechanics.

Technical Skills

Programming: Python (PyTorch, TensorFlow, ASE), Bash, C/C++

Atomistic Simulation: LAMMPS, Classical & Reactive Molecular Dynamics (ReaxFF, REACTER), Gaussian (SCF), ORCA (transition state calculations)

AI & ML: Graph Diffusion Models, GANs, ML Interatomic Potentials (MACE), Model Interpretability

Systems: HPC Clusters, SLURM, GPU Computing, Linux, Git

Conference Publications

- Ahmed, S., Saha, S., Ali, M. A., and Bhattacharjee, A., *"Fabrication of a Cost-Effective Prosthetic Arm Using Electroencephalography Signal,"* International Conference on Industrial & Mechanical Engineering and Operations Management, Dhaka, Bangladesh, December 26–27, 2020.
- Mitra, A., Ahmed, S., Saha, S., and Ali, M. A., *"Fabrication of a Cost-Effective Prosthetic Arm Using Electromyography Signal,"* 5th International Conference on Engineering, Research, Innovation and Education (ICERIE 2019), Sylhet, Bangladesh, January 25–27, 2019.

Projects and Awards in Automation and Robotics

Projects: Developed a brain-controlled, 3D-printed prosthetic arm with EEG-based independent finger control; designed a 4-degree-of-freedom robotic arm for industrial automation; created a maze-solving car with shortest-path algorithm; and built autonomous line-follower and obstacle-avoiding cars.
Awards:

- 2nd Runner-Up, Maze Solver Robot Competition, Robo-Droid Championship, 2016.
- 1st Runner-Up, Battle of Speed, IEEE Robotics Competition, 2015.

Achievements in Mathematics

- Placed in the top three in the National Undergraduate Mathematical Olympiad, organized by the Bangladesh Mathematical Society, in 2014, 2015, and 2017.

Extra-Curricular Activities

Vice President, Bangladeshi Student Association (BSA), 2023–24
Executive Member, BUET Robotics Society, 2017

References

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